PROJECT REPORT: SERVICESHUB

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ROLL NO: FA23-BSE-058

FA23-BSE-004

TECHNOLOGY STACK: FULL-STACK (NODE.JS, EXPRESS, MYSQL)

COURSE: WEB ENGINEERING

SUBMITTED TO: DR. SHAMILA NASREEN

1. Project Title & Introduction

- **Project Title:** ServicesHub – On-Demand Service Booking Platform
- **Introduction:** ServicesHub is a full-stack web application designed to bridge the gap between local service providers (such as Electricians, Plumbers, etc.) and customers. This project leverages modern web technologies like Node.js and MySQL to digitize and simplify the service booking process, making it more efficient for both parties.

2. Problem Statement

Finding reliable local service providers is often a major challenge due to the lack of a centralized system. Most people rely on traditional word-of-mouth methods, which are time-consuming and lack transparency. ServicesHub addresses this issue by providing a dedicated portal where users can access service listings, pricing, and booking status in one place.

3. System Features

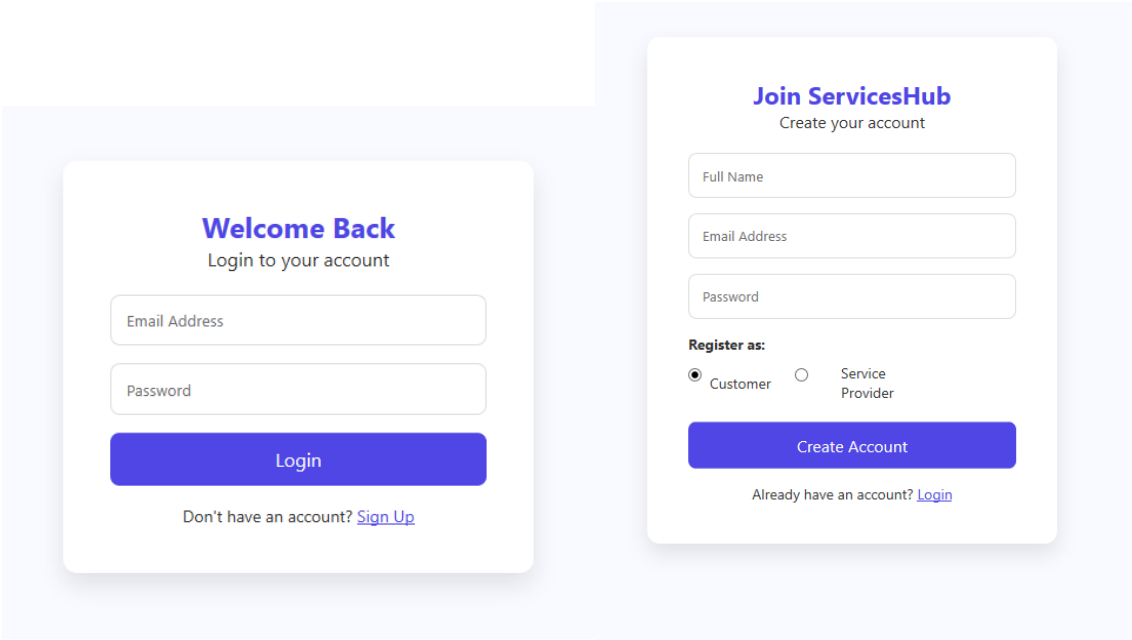
- **User Authentication:** Secure signup and login functionality with role-based access for Customers and Service Providers.
- **Service Catalog:** A dynamic display of all available services along with their descriptions and pricing.
- **Real-time Booking:** An automated system that allows customers to book a service and saves the transaction details directly into the database.
- **Provider Dashboard:** A dedicated interface for providers to view customer bookings, update task status (Update), or cancel/remove records (Delete).
- **Input Validation:** Client-side JavaScript validation to ensure all form fields are correctly filled during registration.

4. Technology Stack

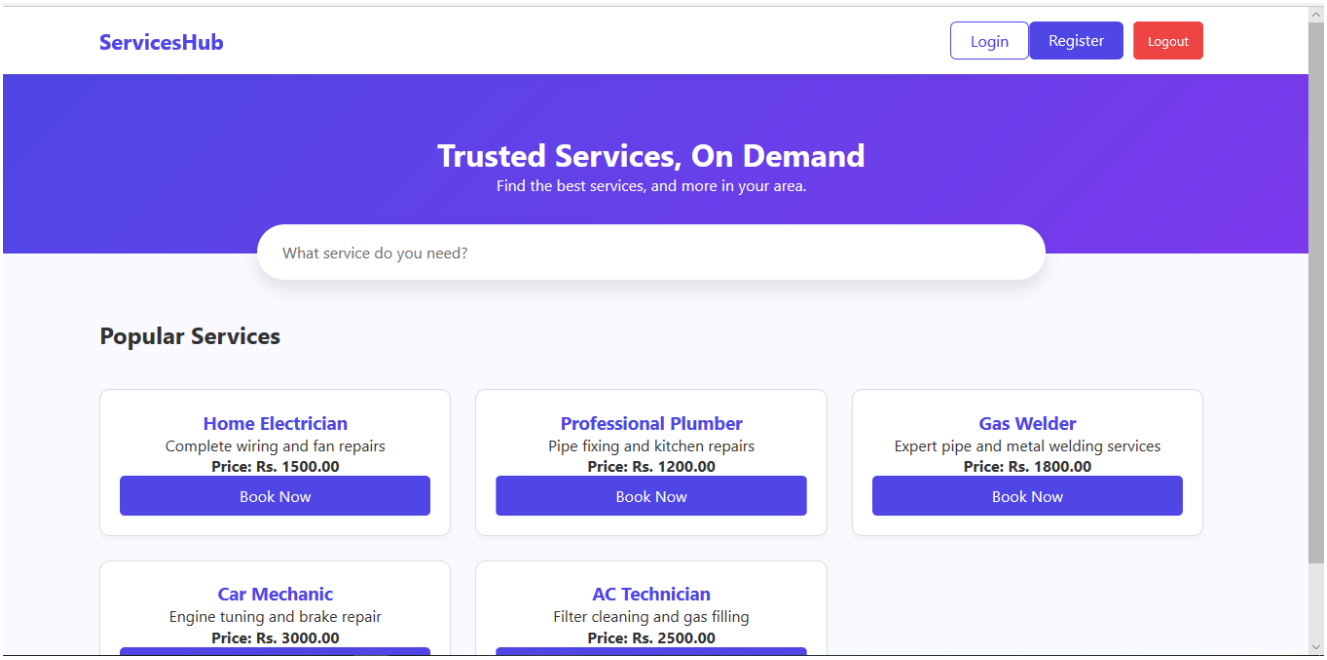
- **Frontend:** HTML5, CSS3, Vanilla JavaScript.
- **Backend:** Node.js, Express.js.
- **Database:** MySQL (Relational Database Management System).
- **Version Control:** Git & GitHub.

5. Frontend Design (Screenshots Required)

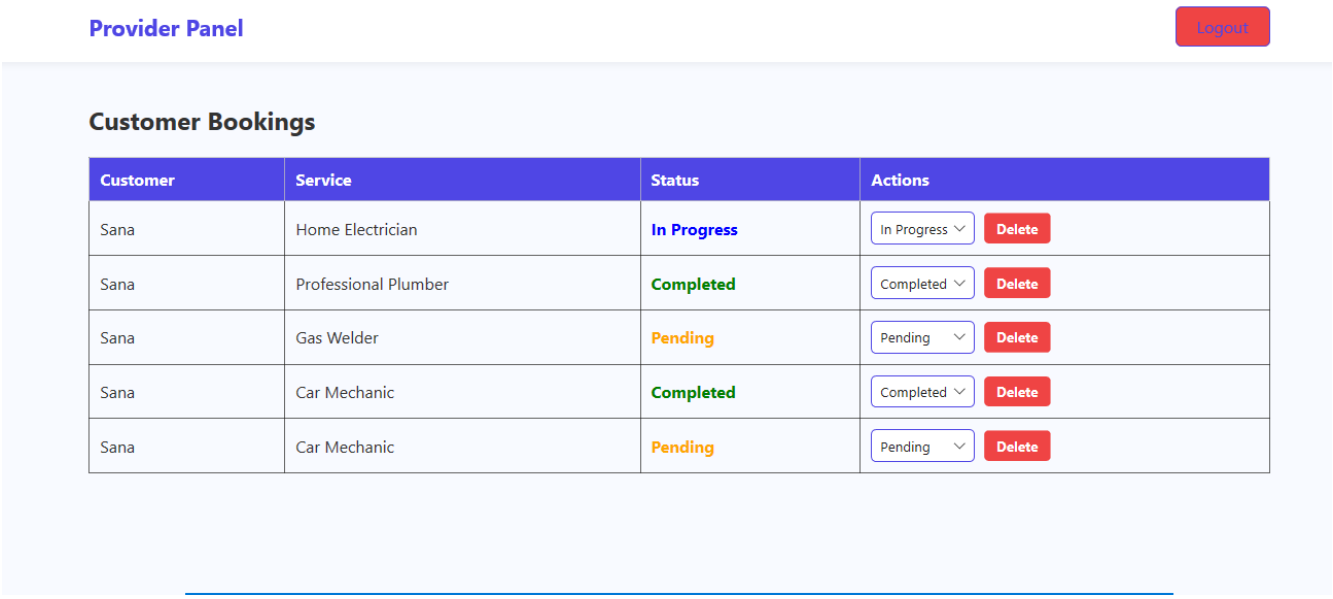
1. **Signup & Login Pages:**



2. **Customer Dashboard:**



3. **Provider Dashboard:**



6. Backend Architecture & CRUD Operations

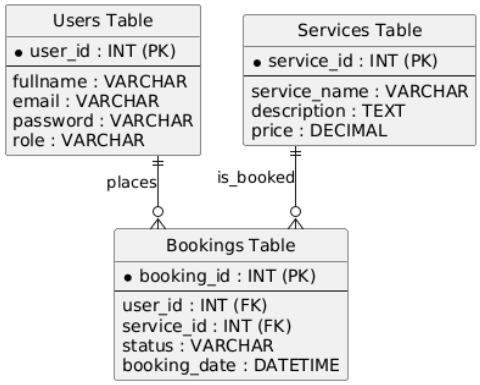
The backend is built on Express.js, which manages the server-side logic and performs the following CRUD operations:

- **Create:** User registration and new booking entries.
- **Read:** Fetching lists of available services and active bookings.
- **Update:** Modifying booking status (e.g., Changing from 'Pending' to 'Completed').
- **Delete:** Removing booking records or canceling service requests from the database.

7. Database Design

The system uses a relational database named `serviceshub_db` consisting of three primary tables:

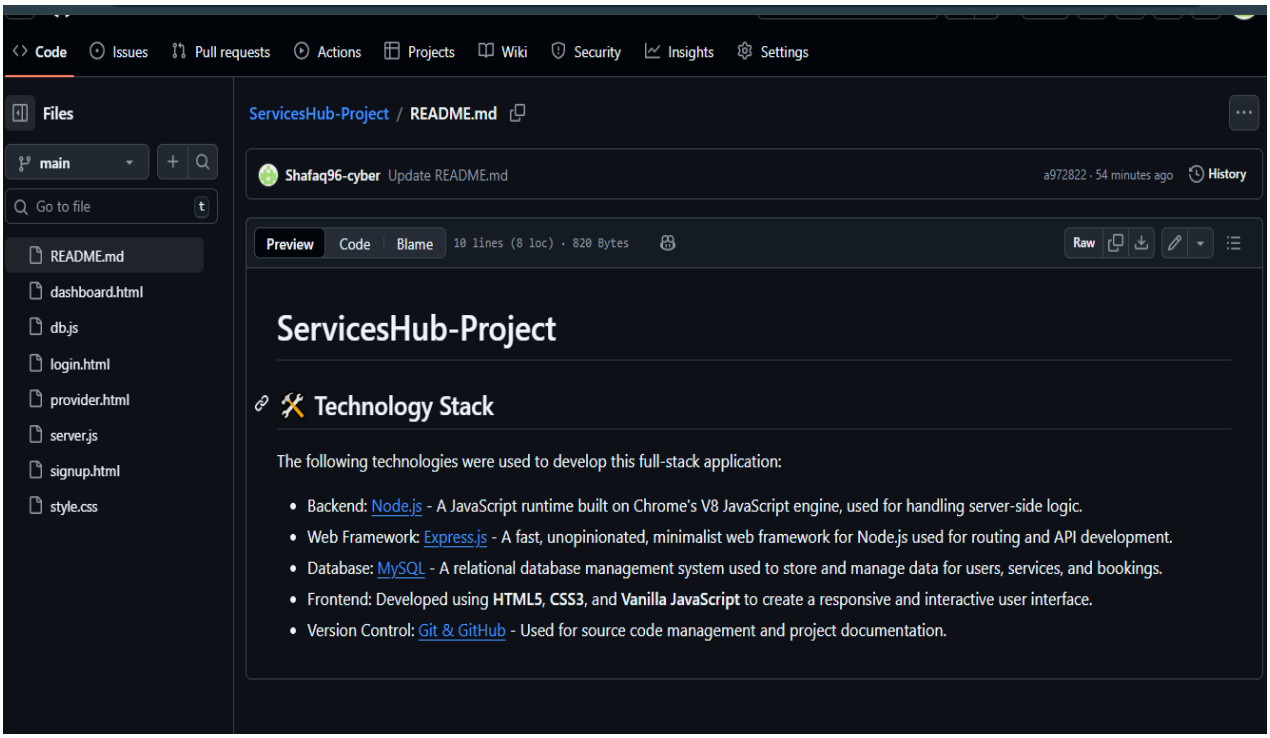
1. **Users Table:** (id, fullname, email, password, role)
2. **Services Table:** (id, service_name, description, price)
3. **Bookings Table:** (id, user_id, service_id, status, booking_date)



8. GitHub Repository

The complete source code is hosted on GitHub to ensure proper version control and documentation.

- **Repository Link:** https://github.com/Shafaq96-cyber/ServicesHub-Project
- **Repository Screenshot:**



9. Conclusion & Future Enhancements

The development of **ServicesHub** has been a significant milestone in understanding the complexities of full-stack web engineering. This project successfully integrates a responsive frontend with a robust Node.js and Express.js backend, supported by a structured MySQL relational database.

Throughout this journey, I have successfully implemented:

- **End-to-End Connectivity:** Establishing a seamless flow between the user interface and the database.
- **Secure CRUD Operations:** Ensuring that data for users, services, and bookings can be created, retrieved, updated, and deleted securely.
- **Real-World Problem Solving:** Creating a digital bridge that solves the communication gap between local service providers and customers.
- **Professional Documentation:** Maintaining high standards of version control through GitHub and technical reporting.

In conclusion, **ServicesHub** is not just a booking portal but a scalable solution that demonstrates how modern web technologies can be utilized to simplify daily life tasks. This project has provided me with a solid foundation in backend architecture, database management, and professional software development practices.